

BRAIN SURVEY

MRI SCANS

The human body is an incredible and complex structure that scientists still seek to properly understand. But it is difficult to study how the living body works without surgery. However, magnetic resonance imaging (MRI) can offer a unique view inside. It produces more detailed scans than X-rays, showing not only bones, but soft tissues as well. MRI machines work by using powerful magnets and radio signals. A computer turns the radio signals into images to create a detailed picture of our insides—even the brain beneath the thick skull.

BRAIN ACTIVITY

In 2015, neuroscientist Rebecca Saxe took this MRI scan of herself and her baby to compare adult and infant brains. She later added the results of another scan to the image, which show the spots in the brain that light up when recognizing faces.



THE HUMAN BRAIN

The brain is a network of billions of cells called neurons that communicate with each other. It is the center of the nervous system, processing information from the senses and controlling muscles. The biggest part, called the cerebrum, is divided into areas called lobes, each specializing in particular functions.

The parietal lobe distinguishes left from right.

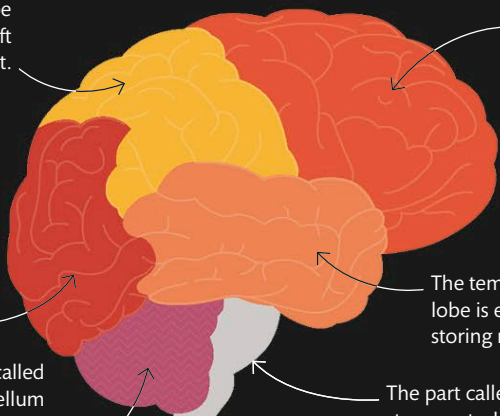
The frontal lobe controls reasoning.

The occipital lobe helps recognize colors.

The temporal lobe is essential for storing memories.

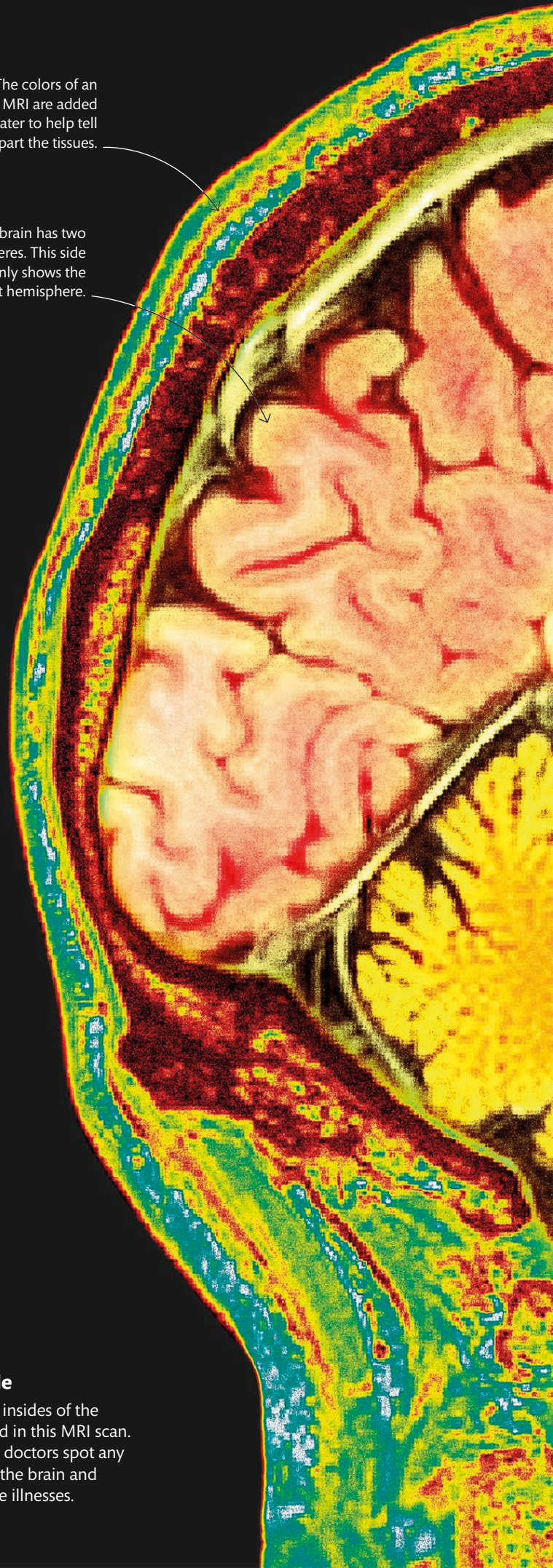
The part called the cerebellum controls balance and coordination.

The part called the brain stem controls breathing.



The colors of an MRI are added later to help tell apart the tissues.

The brain has two hemispheres. This side view only shows the right hemisphere.



Looking inside

The soft, squishy insides of the brain are revealed in this MRI scan. These scans help doctors spot any abnormalities in the brain and diagnose possible illnesses.